Racing Vehicle Lap-Time Optimization with Obstacles and Battery State of Charge

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***Abstract*—This work presents a fast and efficient model predictive control (MPC) framework for lap-time optimization of a 1/10-scale autonomous racing vehicle. The approach integrates three key components: a curvature-based offline maximum-speed profile, a nonlinear Frenet-frame vehicle model with obstacle corridor constraints, and a custom MPC cost designed to maximize progress while maintaining tracking accuracy and smooth control inputs. The offline speed profile is generated using a three-pass algorithm that enforces steady-state cornering, acceleration, and braking limits. The resulting MPC problem is implemented in ACADOS for real-time feasibility. Simulation results on an $8.7\,\mathrm{m}$ compact racing track demonstrate that the controller produces dynamically feasible trajectories, successfully avoids obstacles, and achieves a lap time close to the theoretical minimum obtained from speed profile. The framework provides a computationally efficient foundation for high-speed autonomous racing.**

***Index Terms*—Lap-time, autonomous car, racing track, optimization, NMPC**

I. Introduction

Autonomous racing serves as a demanding benchmark for high-performance control, requiring vehicles to operate near handling limits while minimizing lap time. Model Predictive Control (MPC) is well suited for this task due to its capability to incorporate nonlinear dynamics, constraints, and predictive tracking.

This work adopts the nonlinear vehicle model and obstacle-avoidance strategy from the MPC-RG [1] framework, while omitting its reachability-based robustness layer to maintain computational efficiency. Lap-time performance is further enhanced using an offline curvature-aware maximum-speed profile computed via a standard three-pass method.

The proposed controller integrates (i) the curvature-based velocity profile, (ii) a nonlinear Frenet-frame vehicle model with obstacle corridors, and (iii) a custom MPC cost emphasizing progress, tracking accuracy, and smooth control. Implemented in ACADOS, the approach achieves real-time feasibility.

Simulation results on a 1/10-scale racing track demonstrate accurate path tracking, effective obstacle avoidance, and competitive lap times, validating the effectiveness of the presented framework for fast and efficient autonomous racing.

II. Related Work

A. MPC for Autonomous Racing

Nonlinear MPC (NMPC) has become a dominant approach in racing environments due to its ability to incorporate vehicle constraints, nonlinear tire dynamics, and track boundaries. The MPC-RG framework introduced by [1] extended NMPC for autonomous racing by integrating robust constraint tightening, reachability-based guarding, and a collision-avoidance mechanism driven by predicted lateral error limits. In this project, only the modeling, MPC structure, and obstacle-avoidance submodule of MPC-RG are adopted. The full reachability-based robustness layer is not used. This positions the controller as a deterministic NMPC formulation capable of handling high-speed cornering but without explicit disturbance robustness.

B. Sequential Velocity Profile Generation

Lap-time optimization is strongly dependent on the selection of longitudinal velocity along the track. A widely used strategy is the three-phase curvature-dependent velocity profile approach introduced in [2], in which: Steady-state cornering velocity is computed from the friction limit (see Section V (A)) [2]. Forward integration enforces acceleration feasibility based on propulsion limits. Backward integration enforces braking feasibility based on deceleration limits. This sequential feasibility-based profile generation provides a realistic longitudinal speed reference that respects both tire limits and curvature variations. It has become a de facto standard in modern racing trajectory optimization pipelines and is used in this project to warm-start the MPC and improve numerical stability.

C. Obstacle-Aware Racing and Track Constraints

Obstacle avoidance in racing scenarios commonly relies on separating the track into feasible lateral corridors defined in the Frenet frame. The MPC-RG formulation [1] introduced a practical lateral-error-based obstacle constraint, in which predicted lateral positions are compared with reachable corridor bounds. This method avoids explicit nonlinear distance computations while remaining compatible with real-time optimization. For the present work, the obstacle avoidance scheme from MPC-RG [1] is adopted without the robustness components, enabling efficient collision avoidance using deterministic corridor constraints.

This project integrates these elements into a unified custom-cost MPC formulation aimed at achieving competitive lap-time performance while maintaining feasibility under nonlinear race-car dynamics.

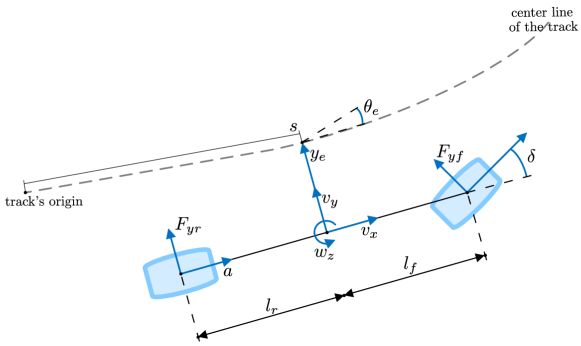

Fig. 1. Standard Bicycle model with curvilinear frame of reference [1]

III. VEHICLE MODELING

Autonomous racing requires a vehicle model that captures nonlinear interactions between longitudinal, lateral, and yaw dynamics while enabling computationally efficient prediction within an MPC framework. This work adopts a nonlinear bicycle model formulated in the track-aligned Frenet frame, Fig. 1 , consistent with the modeling approach used in MPC-RG, but adapted with additional states for progress and battery SoC when needed.

A. Nonlinear Vehicle Dynamics

The state vector is

$$x = \begin{bmatrix} s & v_x & v_y & \omega_z & y_e & \theta_e & \mathrm{SoC} \end{bmatrix}^\top,$$

and the control input is

$$u = \begin{bmatrix} \delta \\ a \end{bmatrix}.$$

The continuous-time nonlinear dynamics are

$$\dot{x} = f(x, u, \kappa(s)),$$

with components detailed below.

1) Planar Dynamics:

$$\dot{v}_x = a + \omega_z v_y - \frac{1}{m}(F_{f,y}\sin\delta - F_{f,x}\cos\delta), \tag{1}$$

$$\dot{v}_y = -\omega_z v_x + \frac{1}{m}(F_{f,y}\cos\delta + F_{f,x}\sin\delta + F_{r,y}), \tag{2}$$

$$\dot{\omega}_z = \frac{1}{I_z}\left(l_f F_{f,y}\cos\delta + l_f F_{f,x}\sin\delta - l_r F_{r,y}\right). \tag{3}$$

2) Tire Forces: The slip angles are

$$\alpha_f = \delta - \frac{v_y + l_f\omega_z}{v_x}, \tag{4}$$

$$\alpha_r = -\frac{v_y - l_r\omega_z}{v_x}. \tag{5}$$

A linear tire model is used:

$$F_{f,y} = C_f\alpha_f, \tag{6}$$

$$F_{r,y} = C_r\alpha_r, \tag{7}$$

and the combined-force friction constraint is

$$F_x^2 + F_y^2 \le (\mu m g)^2.$$

3) Track-Aligned Frenet Dynamics:

$$\dot{s} = \frac{v_x\cos\theta_e - v_y\sin\theta_e}{1-\kappa(s)\,y_e}, \tag{8}$$

$$\dot{y}_e = v_x\sin\theta_e + v_y\cos\theta_e, \tag{9}$$

$$\dot{\theta}_e = \omega_z - \kappa(s)\,\dot{s}. \tag{10}$$

4) Battery Model:

$$\dot{\mathrm{SoC}} = -\frac{1}{C_\alpha}\,p_\alpha, \tag{11}$$

$$p_\alpha = \frac{1}{2}C_d\rho A_r\,v_x^2 \;+\; \mu_s m g\,v_x, \tag{12}$$

Where:

- C_d : aerodynamic drag coefficient,
- A_r : vehicle frontal cross-sectional area (m^2),
- ρ : air density (kg/m^3),
- μ_s : rolling resistance / friction coefficient,
- m : vehicle mass (kg),
- g : gravitational acceleration (m/s^2),
- v_x : longitudinal velocity of the vehicle (m/s),
- C_α : battery capacity (Ah or Coulombs),
- p_α : delivered battery power (W),
- SoC : battery state of charge (0–100%).

IV. MPC FORMULATION

The goal of this work is to design a model predictive controller that drives an autonomous race car around a given track while (i) minimizing lap time, (ii) satisfying physical and safety constraints, and (iii) optionally managing battery state of charge. The problem is posed in continuous time and then approximated via a finite-horizon discrete-time MPC scheme.

The racing MPC designed in this work incorporates a custom objective that combines lap-time minimization, curvature-aware velocity tracking, smooth control actions, and optional SoC management. The formulation is built on the nonlinear dynamics described earlier and solved using ACADOS in real time.

Lap time is minimized by minimizing the longitudinal velocity error with obtained reference longitudinal speed profile,

$$U_x^{\mathrm{ref}}(s)$$

, is computed from the curvature using equation described in section V. The velocity-tracking cost is:

$$\ell_v = q_v\,(v_x - U_x^{\mathrm{ref}}(s))^2.$$

To ensure accurate path tracking:

$$\ell_e = y_e^2 Q_y + \theta_e^2 Q_\theta + v_y^2 Q_{v_y} + \omega_z^2 Q_\omega.$$

These terms keep the vehicle centered on the corridor and aligned with the track direction.

The steering angle δ and acceleration α are regularized to prevent aggressive changes:

$$\ell_u = \delta^2 R_\delta + a^2 R_a.$$

Let

$$\Delta u_k = u_k - u_{k-1}.$$

Then the smoothness cost is:

$$\ell_{\Delta u} = (\Delta\delta)^2 R_{\Delta\delta} + (\Delta a)^2 R_{\Delta a}.$$

Energy optimization is included, a convex penalty is applied:

$$\ell_{\mathrm{SoC}} = q_{\mathrm{SoC}} \left(\mathrm{SoC}_{\mathrm{ref}} - \mathrm{SoC}\right)^2.$$

This term encourages the MPC to maintain enough battery charge for extended racing.

Combining all components, the per-step cost is:

$$\begin{aligned}\ell(x_k, u_k, u_{k-1}) = {} & q_v \left(v_{x,k} - U_x^{\mathrm{ref}}(s_k)\right)^2 + x_k^\top Q x_k \\ & + (u_k - u_{k-1})^\top R_\Delta (u_k - u_{k-1}) \\ & + q_{\mathrm{SoC}} \left(\mathrm{SoC}_{\mathrm{ref}} - \mathrm{SoC}_k\right)^2.\end{aligned}$$

Overall Finite-Horizon Objective: For prediction horizon N, the MPC minimizes:

$$J = \sum_{i=0}^{N-1} \ell(x_{i|k}, u_{i|k}, u_{i-1|k}) + V_f(x_{N|k}),$$

where $\mathcal{V}_f(x)$ is a terminal cost (typically quadratic).

Constraints in the MPC: The full optimal control problem is:

$$\begin{aligned}\min_{\{x_{i|k}, u_{i|k}, SoC_{i|k}\}} \quad & J \\ \text{s.t.} \quad & x_{0|k} = x_k, \\ & x_{i+1|k} = f_d(x_{i|k}, u_{i|k}), \\ & y_{e,i|k}^{\min}(s_{i|k}) \le y_{e,i|k} \le y_{e,i|k}^{\max}(s_{i|k}), \\ & a^{\min} \le a_{i|k} \le a^{\max}, \\ & \delta^{\min} \le \delta_{i|k} \le \delta^{\max}.\end{aligned}$$

V. Simulation Results

This section presents the performance of the proposed MPC controller on the reference racing track. The evaluation includes trajectory tracking, velocity evolution, lateral and heading error, obstacle avoidance, control inputs, and battery state of charge. All simulations were performed in MATLAB R2024b with ACADOS using the nonlinear model described in Section III and the solver configuration summarized in Table IV. The sampling time was set to $T_s = 0.01\,\mathrm{s}$ with a prediction horizon of $N = 50$.

TABLE I
Vehicle Parameters Used in Simulation

Parameter	Symbol	Value
Vehicle mass	m	1.98 kg
Yaw inertia	I_z	0.03 kg m^2
Front axle distance	l_f	0.125 m
Rear axle distance	l_r	0.125 m
Front cornering stiffness	C_f	2.4 N/rad
Rear cornering stiffness	C_r	2.2 N/rad
Friction coefficient	μ	1.2
Gravity	g	9.81 m/s^2
Air density	ρ	1.225 kg/m^3
Drag coefficient	C_d	0.35
Frontal area	A_r	0.05 m^2
Rolling resistance coeff.	μ_s	0.015
Battery capacity	C_α	3000 C

TABLE II
Obstacle Parameters Used in the Simulation

Parameter	Symbol	Value
Obstacle position (curvilinear)	s_{obs}	5.0 m
Obstacle lateral position	$y_{e,\mathrm{obs}}$	0.05 m
Obstacle width	w_{obs}	0.05 m
Obstacle length	L_{obs}	0.05 m
Track width	w_{track}	0.24 m
Safety margin	Δ_{safe}	0.10
Effective corridor bounds	$y_e^{\min}(s)$, $y_e^{\max}(s)$	Computed

A. Offline Velocity Profile Generation

The simulation is performed for lap-time optimization of a 1/10-scale RC electric race car. The experiment uses a closed-loop track of length $8.7\,\mathrm{m}$ containing aggressive maneuvers and varying curvature. Both the curvature $\kappa(s)$ and the track width $w_{\mathrm{track}} = 0.24\,\mathrm{m}$ are known. The maximum curvature magnitude occurs near $s = 4\,\mathrm{m}$. Since the provided curvature data are not continuous, a B-spline interpolation is used to obtain a smooth and differentiable curvature profile suitable for MPC prediction.

To minimize lap time, the controller requires the maximum permissible velocity at each point along the track. Therefore, an offline maximum-speed profile is generated using a standard three-pass algorithm as described in [2]. This algorithm computes the minimum-time velocity profile consistent with tire friction limits and longitudinal force constraints.

The method proceeds in three stages:

1) **Steady-state cornering limit:** The maximum feasible speed at each arc-length position is first computed from the steady-state lateral force limit:

$$U_x^{\mathrm{ss}}(s) = \sqrt{\frac{\mu g}{|\kappa(s)| + \varepsilon}},$$

where μ is the friction coefficient and ε is a small regularization term.

2) **Forward integration (acceleration limit):** A forward pass computes the acceleration-limited velocity using the available longitudinal force $F_{x,\max}$:

$$v_{x,i+1} = \sqrt{v_{x,i}^2 + 2\, a_{\max}\, \Delta s_i}.$$

TABLE III
MPC COST WEIGHTS USED IN THE OPTIMAL CONTROL PROBLEM

Cost Term	Symbol	Value
Progress weight	q_s	1.5
Velocity reference weight	q_v	5.0
Lateral error weight	Q_y	0.5
Heading error weight	Q_θ	0.5
Lateral velocity weight	Q_{v_y}	0.0
Yaw rate weight	Q_ω	0.0
Steering rate weight	$R_{\Delta\delta}$	0.01
Acceleration rate weight	$R_{\Delta a}$	0.01
SoC penalty weight (optional)	q_{SoC}	0–1

TABLE IV
ACADOS SOLVER PARAMETERS USED IN MPC IMPLEMENTATION

Parameter	Symbol	Value
Sampling time	T_s	0.01 s
Prediction horizon	N	50
Integrator type	—	ERK4
Nonlinear solver type	—	SQP
QP solver	—	HPIPM
QP regularization	—	Levenberg–Marquardt
Condensing scheme	—	Partial condensing
Constraint tolerance	—	10^{-6}
Cost tolerance	—	10^{-6}
Max QP iterations	—	20
Warm start enabled	—	Yes
Hessian approximation	—	Gauss–Newton

3) **Backward integration (braking limit):** A backward pass enforces deceleration feasibility:

$$v_{x,i} = \sqrt{v_{x,i+1}^2 + 2\, a_{\min}\, \Delta s_i}.$$

The final offline velocity profile is obtained by taking the minimum across all three limits:

$$U_x^{\mathrm{ref}}(s) = \min\!\left(U_x^{\mathrm{ss}}(s), U_x^{\mathrm{fwd}}(s), U_x^{\mathrm{bwd}}(s)\right).$$

Fig. 2 illustrates the resulting offline velocity profile along the track, which serves as a reference for the MPC controller to improve convergence, stability, and tracking performance.

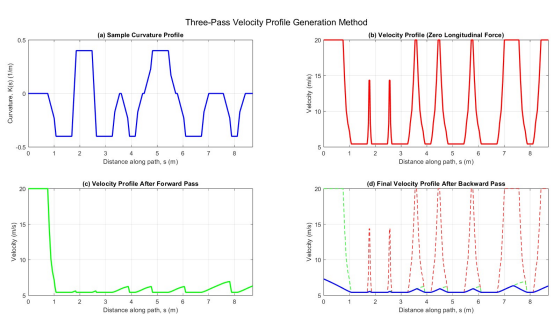

Fig. 2. Offline velocity profile using three pass a) Track curvature b) Velocity Profile (zero longitudinal force) c) Velocity profile after forward pass d) Final velocity after backward pass.

B. Acceleration Along the Track

Fig. 3 shows the distribution of acceleration demands along the track. Higher acceleration regions correspond to straight segments, while reductions in acceleration correlate with high-curvature sections. This behavior matches the curvature-dependent velocity limits, confirming that the MPC respects the friction and track constraints.

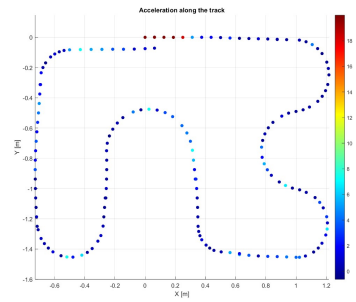

Fig. 3. Acceleration magnitude along the track.

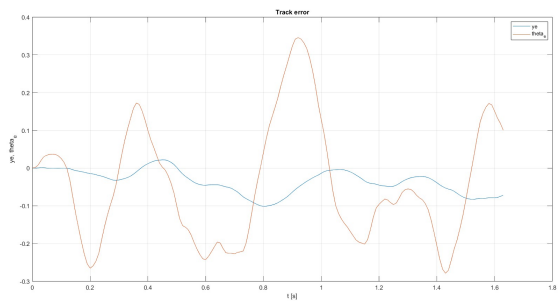

Fig. 4. Lateral error y_e and heading error θ_e over time.

C. Tracking Performance

Fig. 4 illustrates the lateral error y_e and heading error θ_e over time. The controller maintains both errors within tight bounds throughout the entire lap, with deviations increasing slightly in sharp turns. These peaks correspond to curvature spikes where steering demand is highest. The small magnitude of y_e and θ_e demonstrates the controller's ability to stay centered within the track corridor.

D. Obstacle Avoidance and Path Following

Fig. 5 depicts the vehicle trajectory in the presence of an obstacle placed at $s_{\mathrm{obs}} = 5.0\,\mathrm{m}$ with dimensions $0.05\,\mathrm{m}\times 0.05\,\mathrm{m}$. The MPC successfully avoids the obstacle by adjusting its lateral position while staying within the track boundaries. The predicted path smoothly shifts around the obstacle and returns to the optimal line afterward, demonstrating the effectiveness of the corridor-based avoidance strategy.

E. Battery State of Charge

Fig. 6 shows the evolution of the battery state of charge. A slight decrease is observed due to aerodynamic drag and rolling losses, as modeled by the SoC dynamics in Section III. The small SoC consumption over the short simulation horizon validates the physical scaling of the energy model.

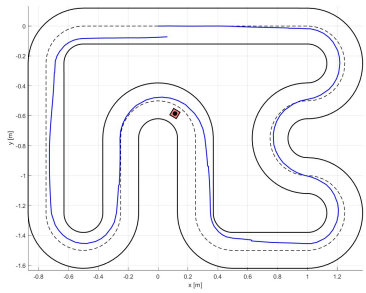

Fig. 5. Vehicle trajectory with obstacle avoidance. The blue line shows the executed path, while the dashed lines denote track centerline.

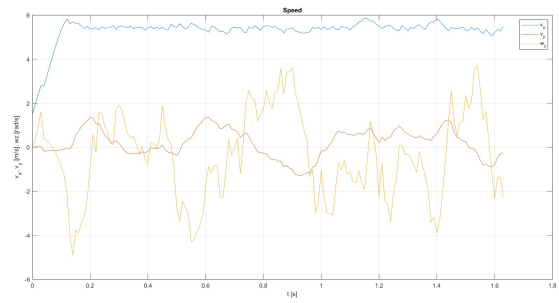

Fig. 7. Vehicle velocities over time: longitudinal v_x, lateral v_y, and yaw rate ω_z.

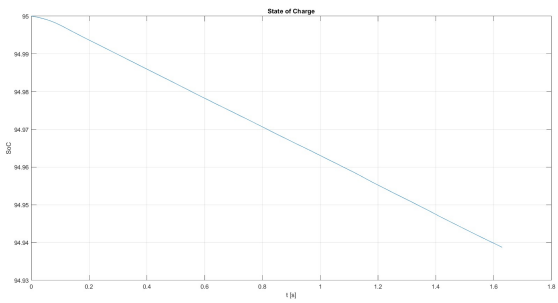

Fig. 6. State of charge evolution during the lap.

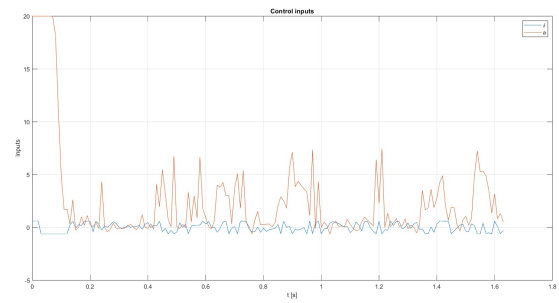

Fig. 8. Control inputs: steering angle δ and acceleration a.

F. Velocity, Lateral Velocity, and Yaw Rate

Fig. 7 presents the evolution of v_x, v_y, and ω_z. The longitudinal velocity v_x rapidly converges to its curvature-limited reference and remains stable throughout most of the trajectory. The maximum velocity obtained from the velocity profile is around 6m/s and MPC was able to keep up with that reference value at each point with some fluctuation due to vehicle dynamics. The lateral velocity v_y and yaw rate ω_z exhibit expected oscillatory behavior during cornering. Peaks in ω_z correspond to tight and wavy track and turns; however, their magnitude remains within physically sensible limits.

G. Control Inputs

Fig. 8 illustrates the applied steering angle and acceleration inputs. The steering input shows sudden variations in acceleration and steering angle within the bounded region due to high acceleration and velocity at sharper changes during high-curvature sections, while the acceleration input varies according to the velocity reference and obstacle avoidance. Despite sudden change, change in control actions is small and within in desired value which reflects the influence of the rate-penalizing terms in the cost function.

H. Computational Performance and Lap-Time Evaluation

Table V summarizes the computational performance of the ACADOS-based MPC solver during the simulation. The average computation time per MPC iteration was $T_c^{\text{avg}} = 8.3 \times 10^{-5}\,\text{s}$, with a maximum time of $T_c^{\max} = 3.39 \times 10^{-3}\,\text{s}$. Both values are significantly below the controller sampling time of $T_s = 0.01\,\text{s}$.

To quantify real-time feasibility, the computation index (CI) is defined as:

$$\text{CI} = \frac{T_c}{T_s}.$$

The average and maximum indices obtained were

$$\text{CI}_{\text{avg}} = 0.0083, \qquad \text{CI}_{\max} = 0.3390,$$

indicating that the solver consistently completed within the allowed sampling interval. Since $\text{CI}_{\max} < 1$, the controller satisfies real-time constraints.

The minimum theoretical lap time computed from the curvature-based velocity profile was $t_{\text{theoretical}} = 1.5373\,\text{s}$. The MPC achieved a lap time of $t_{\text{MPC}} = 1.65142\,\text{s}$, representing an 8.0% increase relative to the theoretical lower bound. This difference is primarily attributed to obstacle avoidance, lateral-error corrections, and control. Nevertheless, the achieved lap time demonstrates competitive performance within dynamic and safety constraints.

I. Discussion

Overall, the results demonstrate that the custom-cost MPC achieves stable trajectory tracking, efficient obstacle avoidance, and smooth control behavior. The controller respects velocity limits imposed by curvature, maintains low lateral and heading error, and handles obstacles without requiring

TABLE V
MPC TIMING AND LAP-TIME PERFORMANCE

Metric	**Value**
Average MPC solve time T_c^{avg}	$8.3 \times 10^{-5}\,\text{s}$
Maximum MPC solve time $T_c^{\max}$	$3.39 \times 10^{-3}\,\text{s}$
Sampling time T_s	$0.01\,\text{s}$
Computation index (avg.) CI_{avg}	0.0083
Computation index (max.) $\text{CI}_{\max}$	0.3390
Theoretical lap time	$1.5373\,\text{s}$
Achieved lap time (MPC)	$1.65142\,\text{s}$
Real-time status	REAL-TIME OK

reachability-based robustness. These outcomes validate the effectiveness of the proposed approach in fast lap-time scenarios while maintaining feasibility and safety constraints.

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